CYBER SECURITY PRACTICAL REPORT

# IP Address Blocking Using UFW on Kali Linux

Source-based firewall filtering and cross-VM connectivity testing

| | |
|---|---|
| **Firewall host** | Kali Linux |
| **Firewall utility** | UFW (Uncomplicated Firewall) |
| **Test client** | Windows 10 x64 virtual machine |
| **Kali IP address** | 192.168.182.128 |
| **Windows IP address** | 192.168.182.129 |
| **IP-blocking rule** | Deny inbound traffic from 192.168.182.129 |
| **Environment** | VMware Workstation virtual network |

This report documents the UFW source-IP deny rule and the Windows-to-Kali connectivity tests shown in the supplied screenshots. Results are reported as observed; inconsistent test outcomes are identified rather than presented as conclusive proof of blocking.

# 1. Objective

Configure UFW on Kali Linux to deny inbound traffic originating from a selected IPv4 address, then attempt network connections from that source device to the Kali host.

# 2. Lab Topology and Addressing

| System | Role | IPv4 address |
|---|---|---|
| Kali Linux | Firewall / destination | 192.168.182.128 |
| Windows 10 | Source / test client | 192.168.182.129 |

# 3. Configuration Command

The following command was executed on Kali Linux to add a rule denying inbound traffic from the Windows VM's IP address:

```
sudo ufw deny from 192.168.182.129
```

The terminal returned **Rule added**. The subsequent numbered UFW status output showed an entry equivalent to **Anywhere DENY IN from 192.168.182.129**. This is a source-address rule and is intended to apply to inbound traffic from that IPv4 address, across ports.

# 4. Screenshot Evidence – Kali Linux

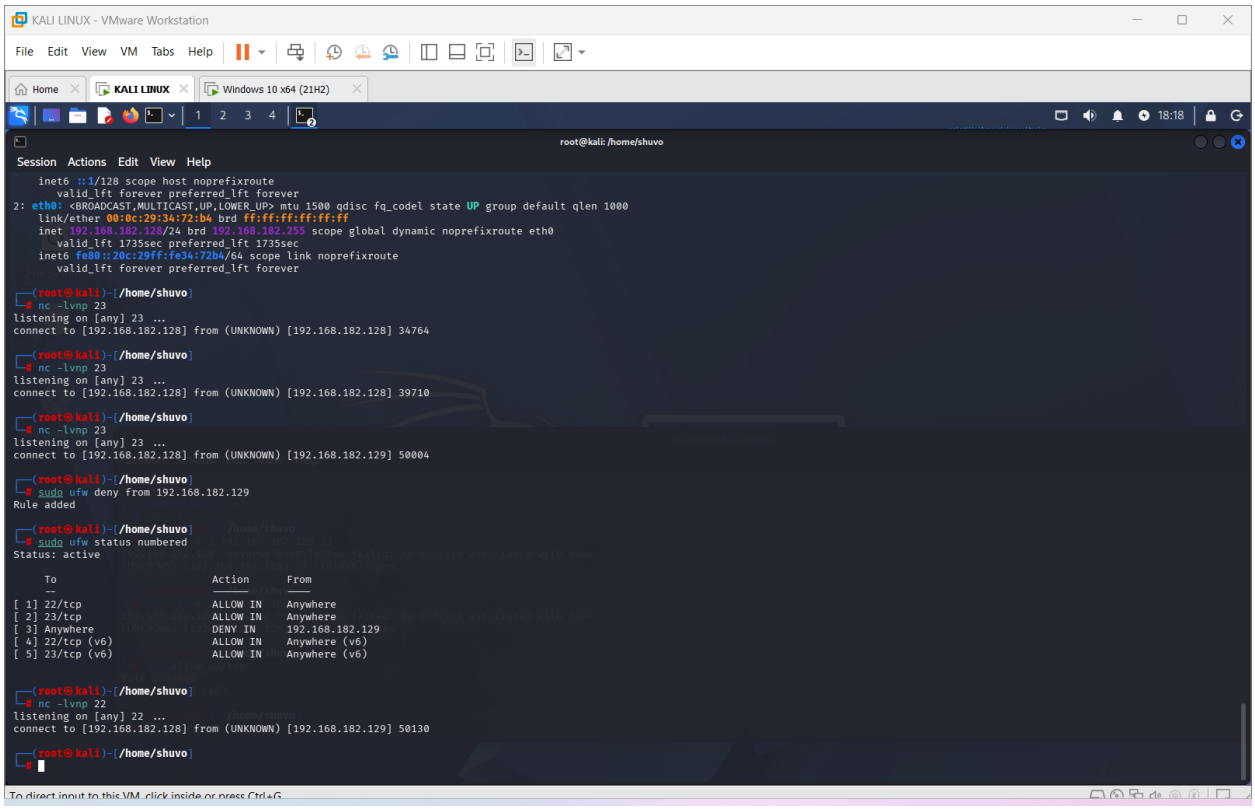



*Figure 1. Kali Linux terminal showing the source-IP deny command, the numbered UFW rules, and a Netcat listener on TCP port 22. The displayed rule list includes the deny rule for 192.168.182.129.*

# 5. Windows Connectivity Tests

Windows PowerShell was used to test TCP connectivity to the Kali host at 192.168.182.128. The screenshots show tests against ports 23 and 22, with Windows reporting source address 192.168.182.129.

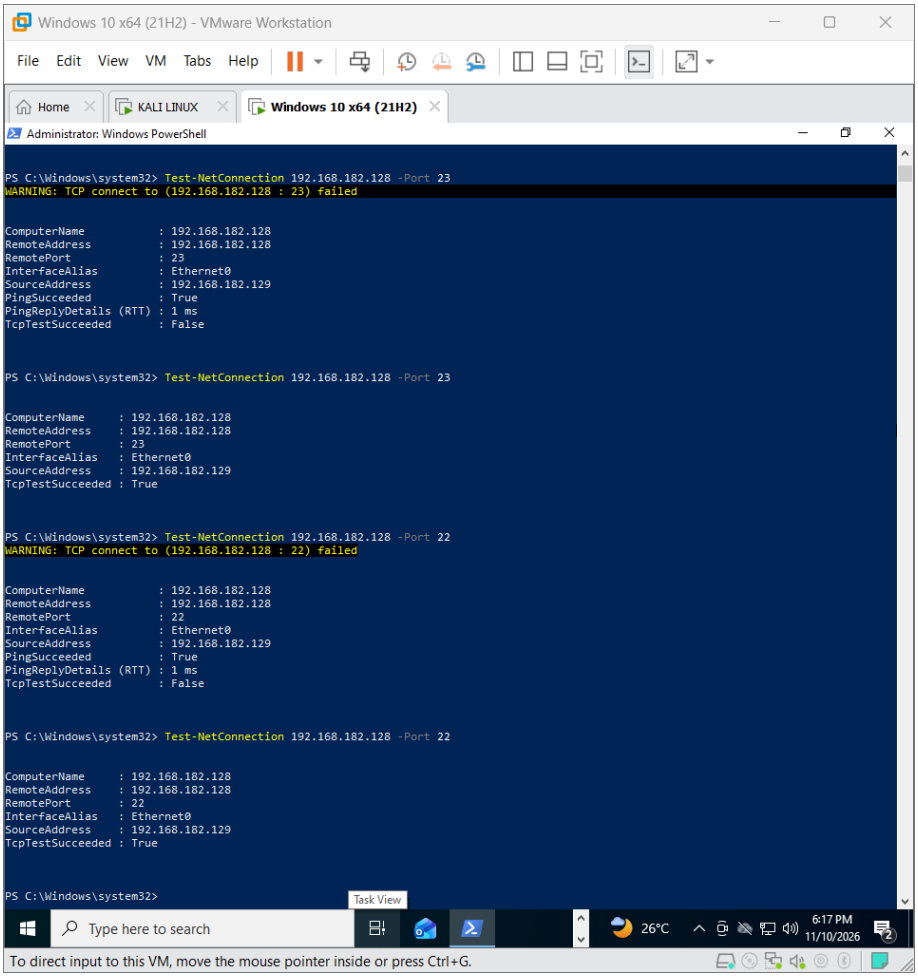



*Figure 2. First Windows test screenshot. It shows TCP/23 failing in one test and succeeding in a later test; TCP/22 also shows a failed test. These results vary between attempts.*

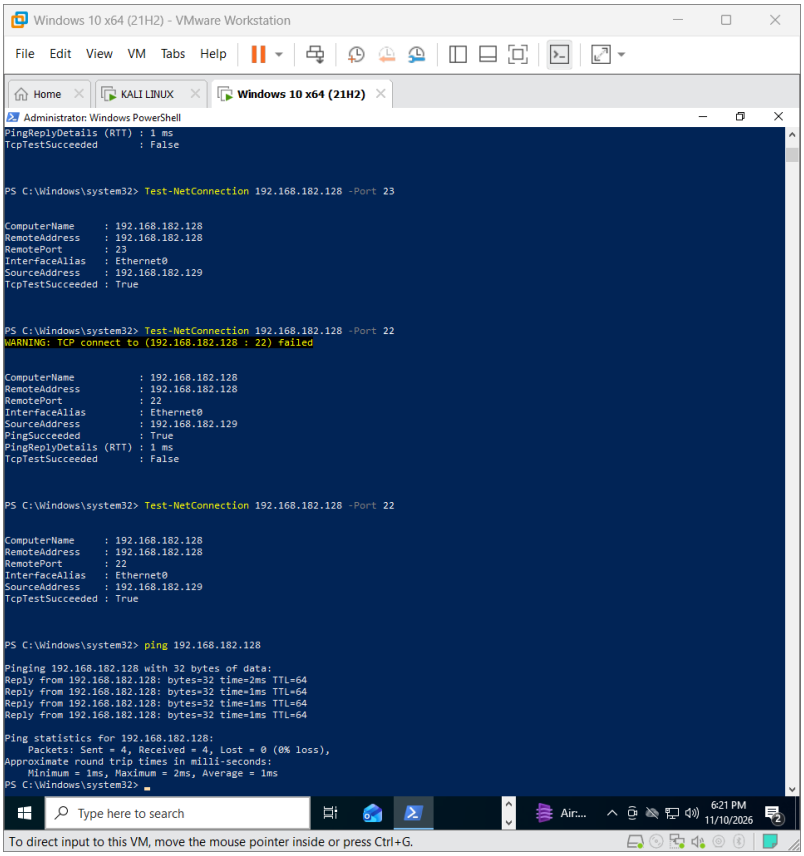



*Figure 3. Later Windows tests. TCP/23 succeeds, TCP/22 first fails and then succeeds, and ICMP ping to the Kali host succeeds.*

# 6. Results and Technical Interpretation

| Evidence | Observed output | Meaning |
|---|---|---|
| UFW source rule | DENY IN from 192.168.182.129 | Rule was added and displayed by UFW. |
| Windows → Kali TCP/23 | False in one attempt; True in a later attempt | Port test was inconsistent. |
| Windows → Kali TCP/22 | False in one attempt; True in a later attempt | Port test was inconsistent. |
| Windows → Kali ping | Replies received; 0% packet loss | Kali host remained reachable by ICMP. |
| Kali Netcat on TCP/22 | Listener displayed a connection from 192.168.182.129 | A connection from the Windows source was observed in the Kali ter |

**Important finding:** The screenshots confirm that the UFW source-IP deny rule was added. However, the Windows tests are not consistently blocked: later screenshots show successful TCP connections to ports 23 and 22, and the Kali Netcat terminal shows a connection from 192.168.182.129 to port 22. Therefore, the supplied evidence does **not** conclusively demonstrate that the IP-block rule successfully blocked all traffic. The rule configuration is evidenced, but its enforcement needs a repeatable retest.

# 7. Recommended Retest Procedure

To obtain conclusive evidence, first stop all Netcat listeners with Ctrl+C and make sure no other process is bound to the test ports. On Kali, verify that UFW is active and inspect the complete rule order:

```
sudo ufw status numbered
sudo ufw status verbose
```

Remove any duplicate or conflicting allow rules for the test service if appropriate, then add the source-IP deny rule again. UFW normally evaluates its rules in order; inspect numbered rules and use a specific insertion position if an earlier allow rule is taking precedence. For a controlled test, insert the deny rule at the top:

```
sudo ufw insert 1 deny from 192.168.182.129
```

Next, run a known test service on Kali, then test from Windows using Test-NetConnection. Capture the Kali rule list and the Windows result at the same time. Confirm the actual packet path and rule counters/logs if results remain inconsistent. Do not treat ping alone as proof of TCP filtering.

# 8. Cleanup – Remove the Test Rule

List the rules and delete the exact source-IP deny rule by its current number:

```
sudo ufw status numbered
sudo ufw delete RULE_NUMBER
```

Replace RULE_NUMBER with the number shown for the source-IP deny rule. Recheck the rule list afterward. If you want to reset all UFW rules instead, use **sudo ufw reset** and confirm the prompt; this resets UFW configuration and disables it.

# 9. Conclusion

A source-IP deny rule for 192.168.182.129 was successfully added and displayed in UFW on Kali Linux. The supplied screenshots also record connectivity tests from Windows 10. Because some later TCP tests succeeded and a connection from the Windows IP was displayed by Netcat, the practical should be recorded as **rule configured; enforcement not yet conclusively verified**. A clean retest with rule ordering checked is required before claiming successful IP blocking.

# 10. Submission Checklist

| Item | Status |
|---|---|
| UFW source-IP deny command | Captured |
| Numbered UFW rule list | Captured |
| Windows source IP and destination IP | Captured |

| | |
|---|---|
| Connectivity test screenshots | Captured – inconsistent results |
| Successful enforcement verification | Retest required |
| Cleanup instructions | Included |

Prepared from the three screenshots supplied for this practical. Add the final verified retest screenshot and your GitHub repository URL before submission.